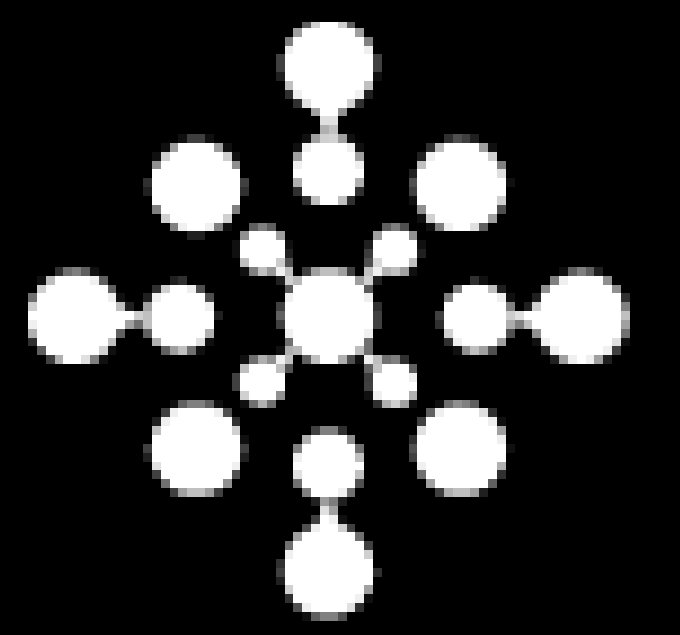




# What is Visualization for Communication? Analyzing Four Years of VisComm Papers

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## INTRODUCTION

Although visualizations for communication are some of our earliest examples of visualization, visualization research shifted away from studying visualization purely as a means of communication. As a result, there are few theories or guidelines specific to visualizations for communication. To address this gap, we surveyed VisComm workshop proceedings from 2018 to 2022. We looked at how papers define visualization, sorted them into four categories, and identified design themes within each. We found accessibility and user-centric design to be common themes across papers. However, despite this emphasis, we found a definitive lack of concrete remarks on incorporating Diversity, Equity, and Inclusion (DEI) principles into visualization for communication designs.

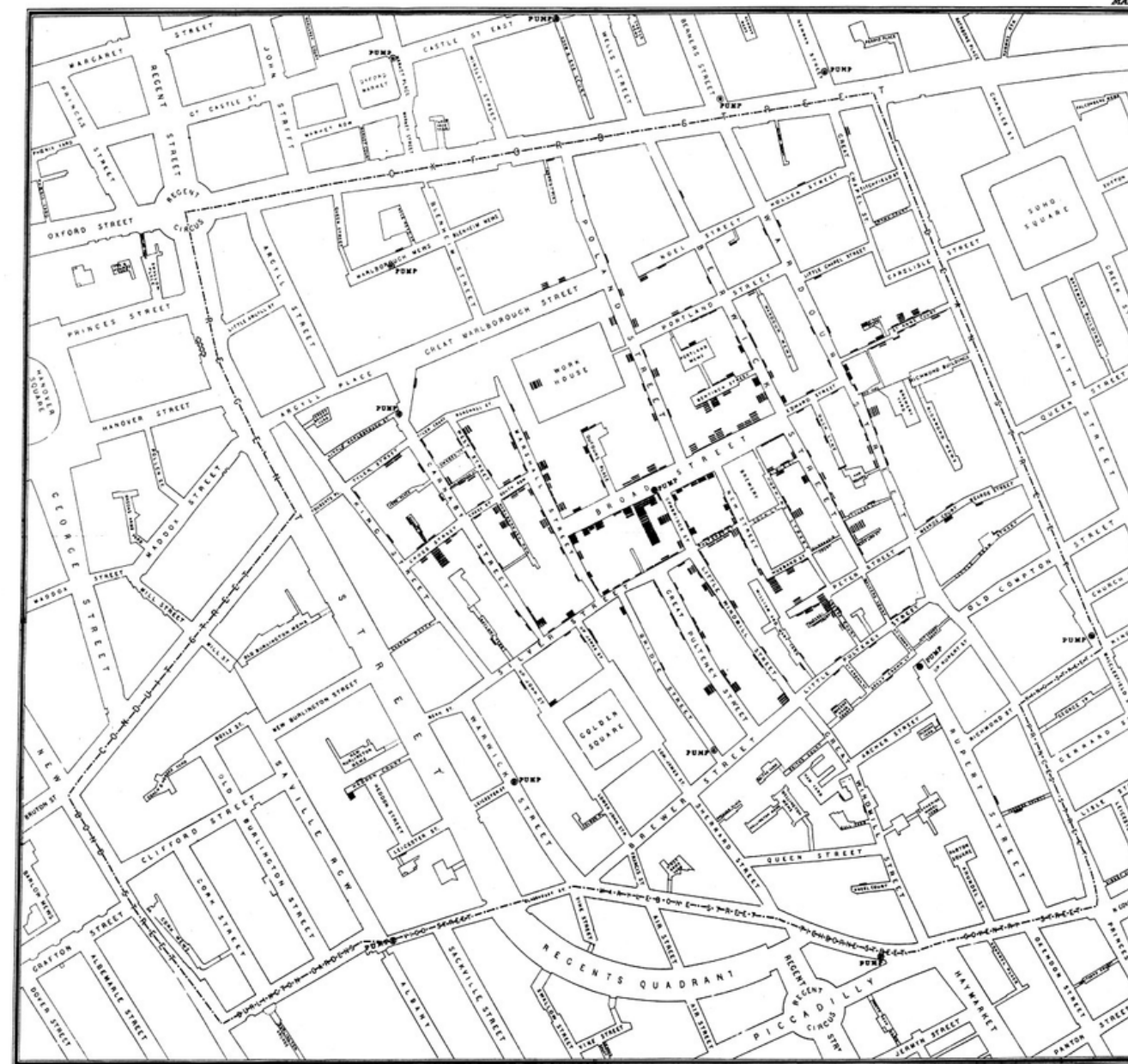
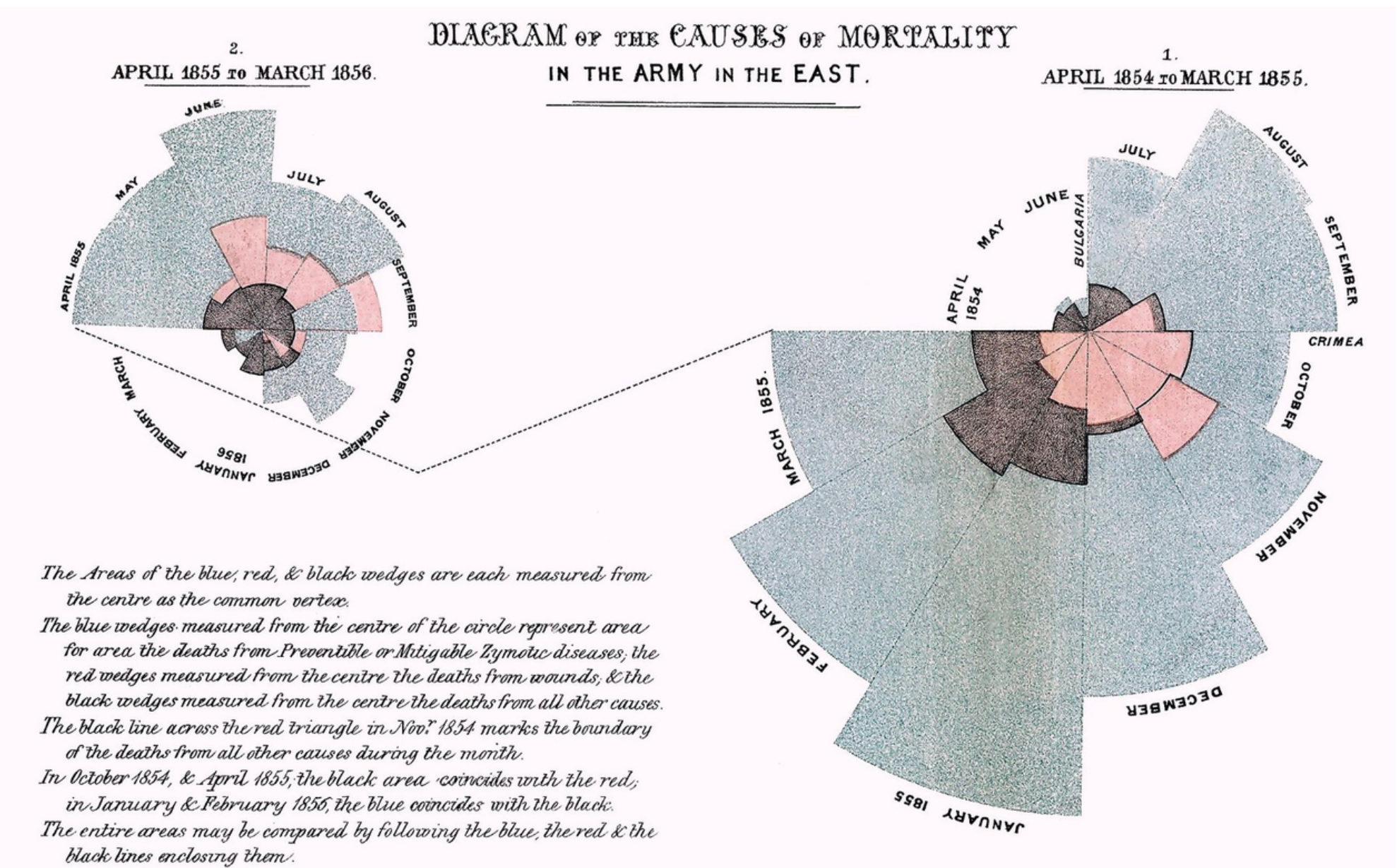
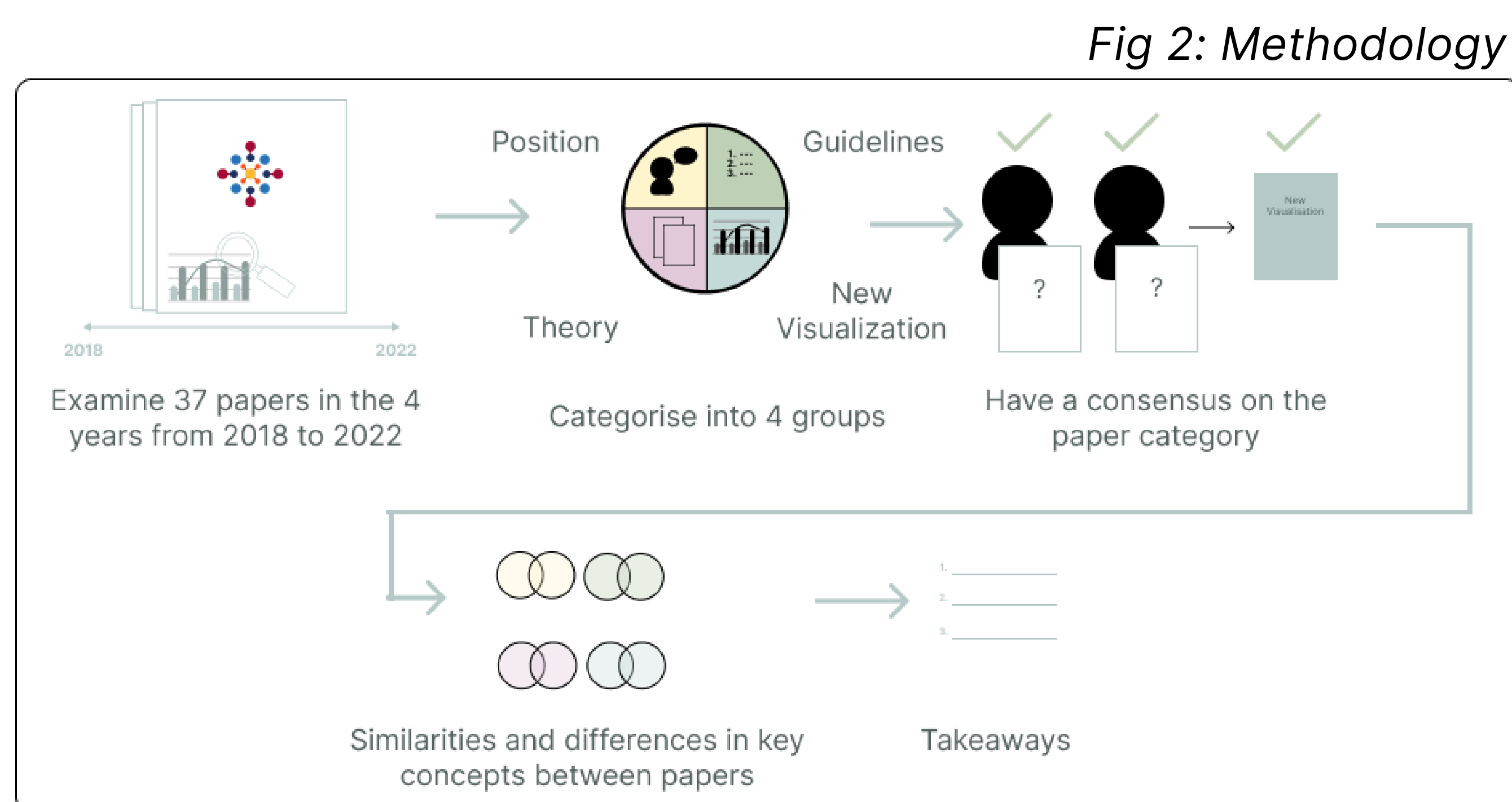


Fig 1: John Snow's Cholera Map and Florence Nightengale's Rose Diagram



## METHODOLOGY



Papers were categorized into four groups:

- **Guideline:** papers offering design guidelines,
- **New Visualization:** papers introducing novel visualization,
- **Position:** papers expressing subjective viewpoints, and
- **Theory:** papers proposing theoretical frameworks.

Applying grounded theory principles, researchers analyzed:

- (a) how papers defined visualization and
- (b) design takeaways within each group.

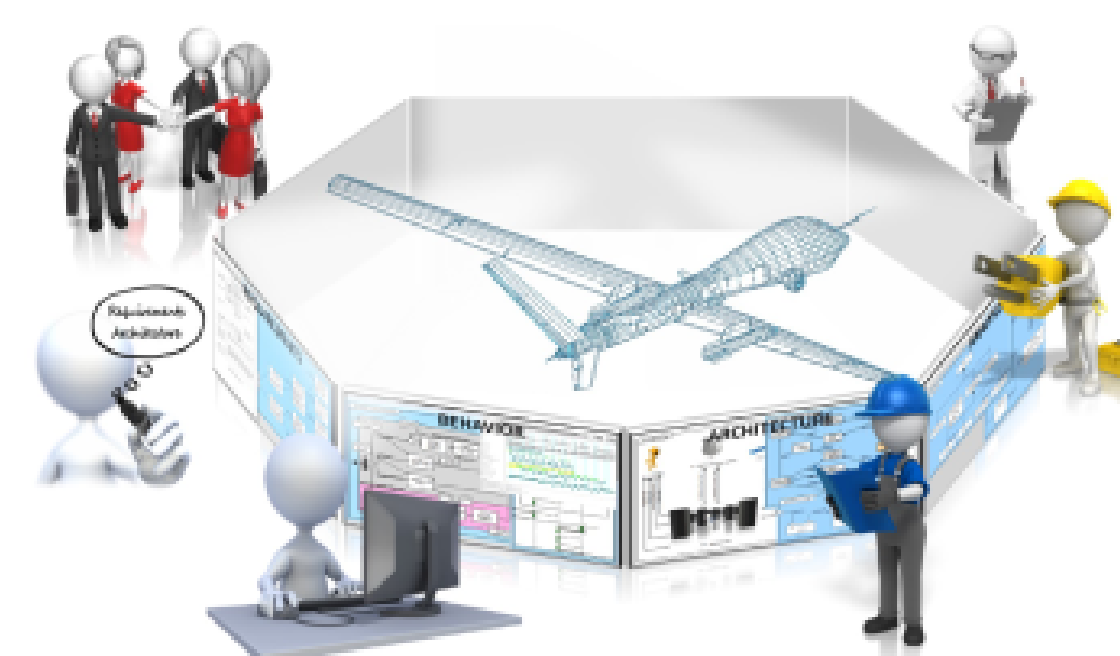
## DEFINITIONS

We found definitions of visualization within VisComm proceedings emphasize **centering the audience, communication, and broad understandability.**



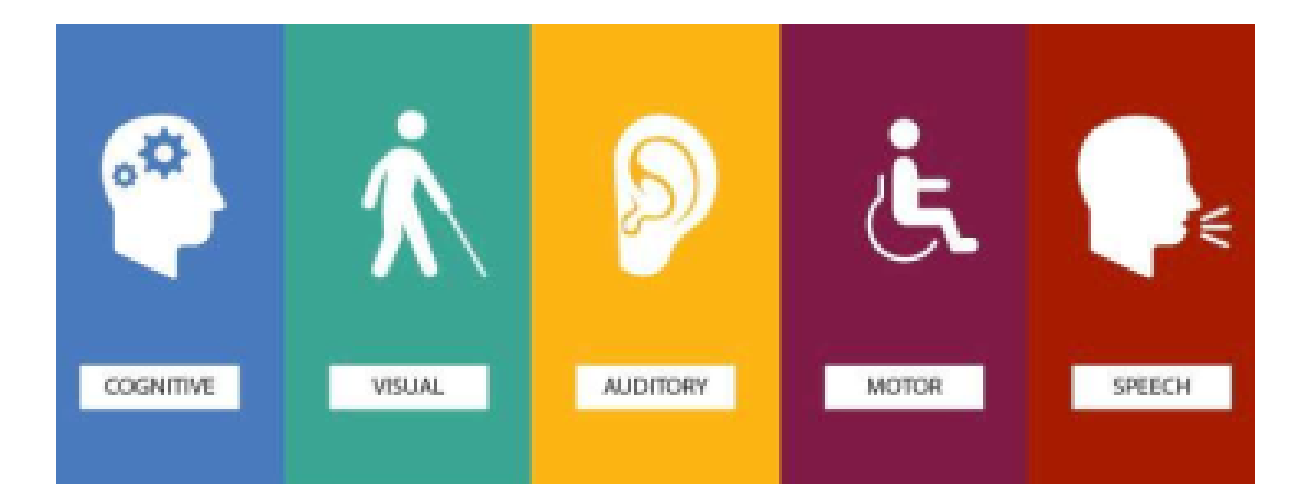
**Diverse Audience**  
Encompass wide audience and aiming for widespread engagement

MHPTeamSI - How to Authentically Diversify Your Marketing Strategies



**Communication Tool**  
Encompass wide audience and aiming for widespread engagement

VITech Communication for Visualization



**Accessible**  
Encompass wide audience and aiming for widespread engagement

Argcis - building an accessible product our journey so far

Fig 3: Characteristics of VisComm's definition of visualization

## TAKEAWAYS

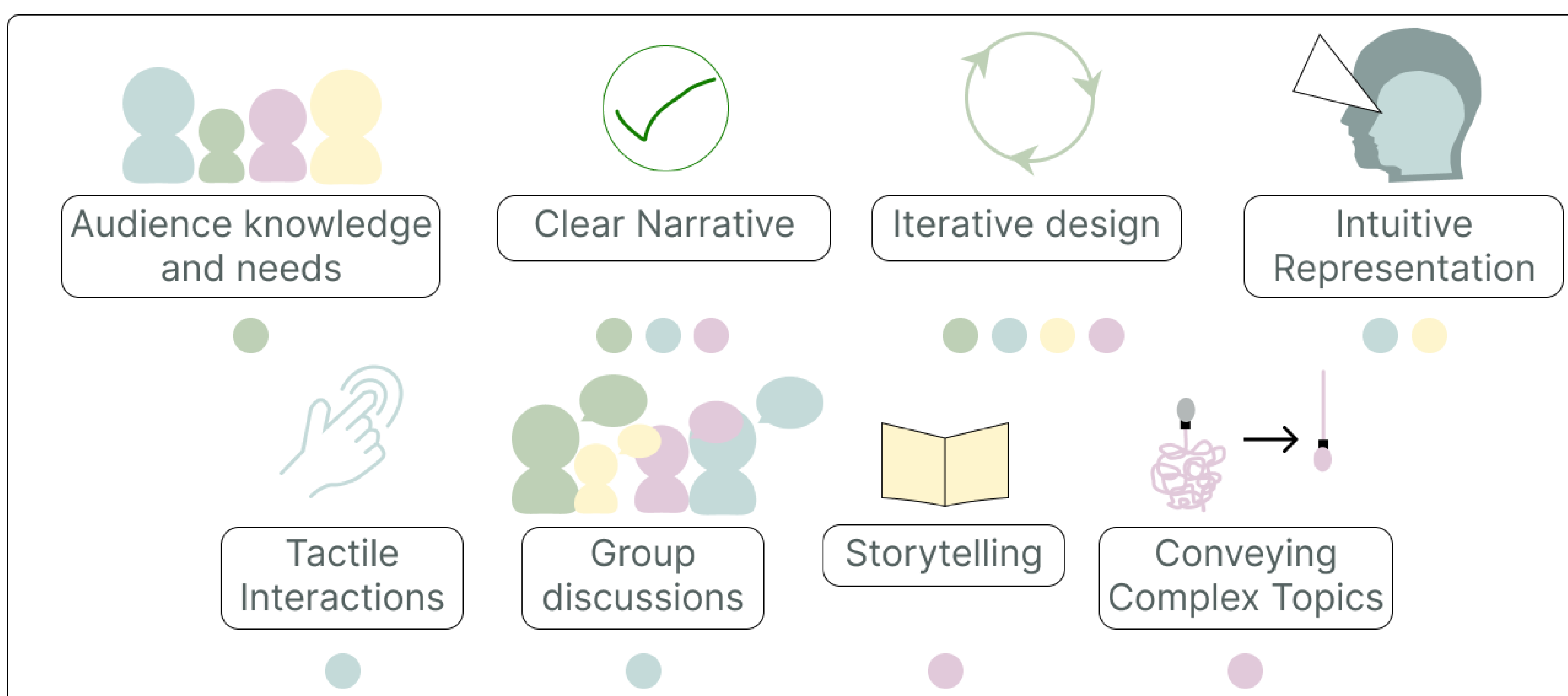
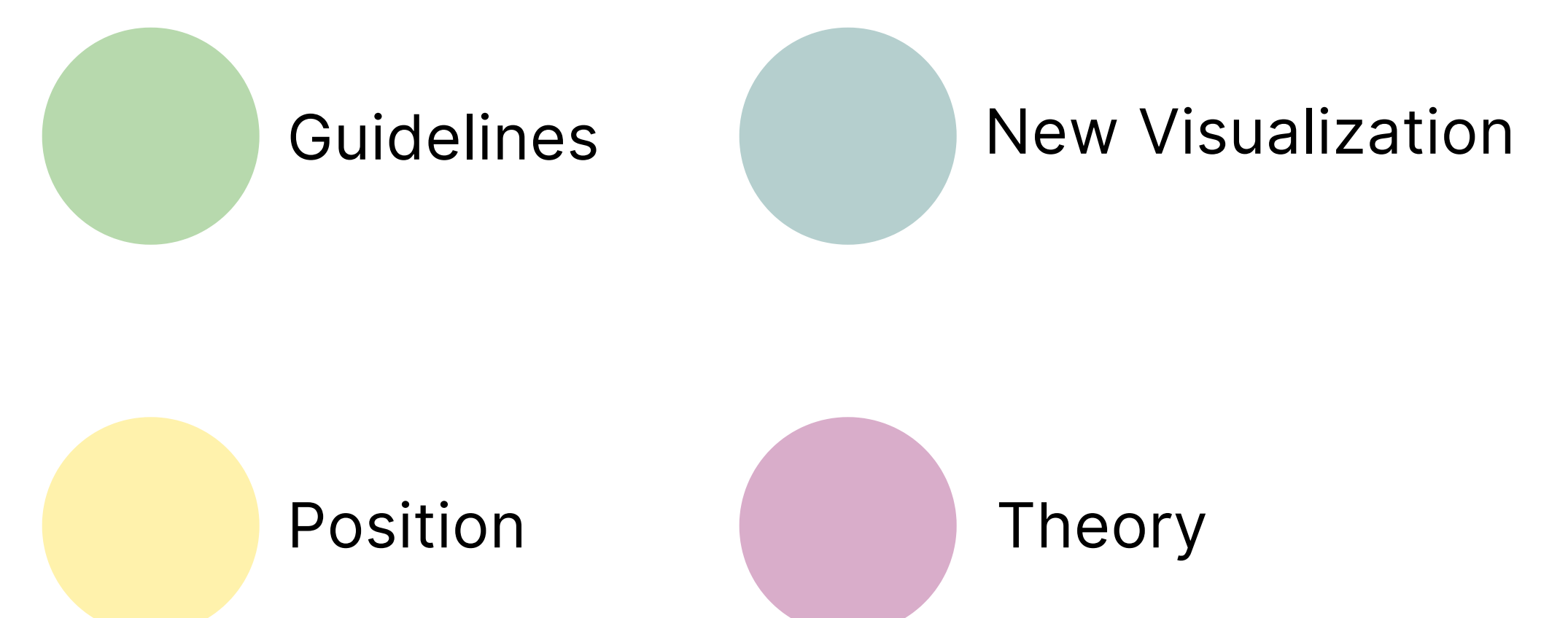


Fig 4: Key design themes between categories

Recurring themes of iterative design and user consideration were observed across different categories, reflecting the overarching emphasis on audience-centric visualization within this domain.



## CONCLUSION

Most of the themes in each paper group aligned with VisComm's definition of visualization. However, a lack of discussion on implementing and designing inclusive and accessible visualizations was observed. In fact, only 3 of the 37 papers surveyed explicitly mentioned inclusive design. Our hope is that this preliminary analysis of VisComm work motivates visualization for communication researchers to pay explicit attention to DEI in their research and practice.